Laboratory Report

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Section: BSIT 302i

1. Which columns contain missing values, and how did you handle them?

The columns that contained missing values were mainly Runtime (Minutes) and some categorical information such as Genre and Description. Missing numerical values were replaced with 0, while missing categorical or text values were given an appropriate label such as Unknown or Not Available. Blank and unnecessary records were also removed when necessary.

2. How did you identify and correct duplicate records?

I used Conditional Formatting to highlight duplicate movie titles. After identifying the duplicates, I used the Remove Duplicates function under the Table Design menu and selected the Movie Title column to remove duplicate records.

3. How did you correct inconsistent data?

I used Find and Replace to correct spelling and category inconsistencies. For example, `Sci-Fi` was changed to Science Fiction, and misspelled categories were corrected. I also used the TRIM function to remove unnecessary spaces from movie titles and converted the column headings to uppercase for consistency.

4. What Excel functions did you use in the data analysis?

I used several Excel functions, including:

- `SUM()` – to calculate the total budget
- `AVERAGE()` – to calculate the average audience rating
- `MIN()` – to find the minimum budget
- `MAX()` – to find the maximum box office
- `COUNT()` – to count numerical records
- `COUNTIF()` – to count records meeting a condition
- `SUMIF()` – to calculate totals based on a condition
- `AVERAGEIF()` – to calculate averages based on a condition
- `TRIM()` – to remove extra spaces
- `UPPER()` – to convert headings to uppercase

5. What did you learn from this laboratory exercise?

I learned how important data cleaning is before performing data analysis. Inconsistent spelling, extra spaces, missing values, and duplicate records can affect the accuracy of the results. I also learned how to use Excel functions, conditional formatting, tables, and charts to organize, analyze, and present data more effectively.

6. What is the importance of data visualization?

Data visualization makes information easier to understand and compare. The charts in the Dashboard helped me visualize the **box office performance, audience ratings, and movie runtimes**. Instead of looking only at numbers in a table, charts make patterns and differences easier to identify.

Conclusion

This laboratory exercise helped me understand the basic process of data cleaning, analysis, and visualization using Microsoft Excel. I learned how to prepare an unorganized dataset, correct errors, remove unnecessary records, use formulas for analysis, and create charts for better presentation. These skills are useful for turning raw data into meaningful and understandable information.